Numerical approximation of the two-component Hunter–Saxton system on moving and fixed meshes

Abstract

Integrable semi-discretizations provide exact wave solutions on evolving spatial lattices, allowing the numerical evolution and the approximation of continuous waves to be examined separately. We use this connection to study the two-component Hunter–Saxton system on moving and fixed meshes. Starting from an existing integrable semi-discretization, we formulate a coupled evolution of the velocity differences and mesh intervals and compare it with a direct moving-grid approximation and centered finite differences on a fixed physical grid. An explicit identity between the two moving-grid vector fields identifies a first-order finite-lattice correction. Exact one- and two-soliton solutions then allow this correction to be examined alongside the errors introduced by time integration and physical-coordinate reconstruction. Comparisons of forward Euler, Runge–Kutta and Crank–Nicolson integration distinguish accurate tracking of a semi-discrete trajectory from accurate approximation of the continuous fields. The density conservation law also explains the geometry of the natural mesh: for the smooth solitons considered here, the mesh expands near velocity peaks. We construct two further conserved coordinates by balancing curvature against mass stretching and by cancelling slope–density exchange in a quadratic balance. The three meshes retain approximately second-order refinement with different fieldwise error coefficients. The curvature–mass ratio improves density accuracy in both soliton tests and velocity accuracy for the two-wave profile; the energy-type density further reduces the single-wave density error while increasing the velocity error. Finally, independent numerical evolutions with and without a prescribed velocity perturbation reveal its persistence and transfer into the density field. Overall, in this study we have examined the effects of spatial approximation and time integration separately, and explored how field reconstruction, mesh allocation and coupling shape the accuracy and response of the numerical solutions. These comparisons show how the analytical structure of the 2HS system can help explain the behavior of its numerical approximations and inform the choices made in their construction.

Keywords: two-component Hunter–Saxton equation; integrable semi-discretization; reciprocal transformation; moving mesh; numerical error; perturbation response

1. Introduction

Nonlinear wave equations offer a setting in which coherent structures can be followed through both analytical formulas and numerical evolution. The Hunter–Saxton equation originated as an asymptotic model for director-field dynamics in nematic liquid crystals [1]. Its two-component extension couples a velocity field to a transported density and admits a bilinear description through a hodograph transformation. Lou, Feng and Yao constructed a general multi-soliton solution for the two-component Hunter–Saxton (2HS) equation in this framework [2]. The density therefore has two roles in the computation: it is a physical field to be resolved and a quantity that determines the relation between computational and physical coordinates.

Hori, Tanaka, Maruno and Ohta developed an integrable semi-discretization of the 2HS equation using bilinear equations, determinant solutions and a discrete hodograph transformation [3]. Their construction supplies both a spatial model and exact finite-lattice trajectories. A related numerical approach was developed by Feng, Sheng and Yu for a generalized sine-Gordon equation [4], where integrable and non-integrable semi-discrete systems generate self-adaptive moving-mesh methods. These constructions give a concrete way to use analytical solution structure in numerical evolution: the field values and the physical node positions evolve together.

Numerical studies of Hunter–Saxton-type equations have also used their geometric and conservation properties to guide discretization. Miyatake, Cohen, Furihata and Matsuo developed multi-symplectic and

Hamiltonian-preserving methods for the scalar, modified and two-component equations, with particular attention to the boundary conditions required by the numerical formulations [5]. For the scalar Hunter–Saxton equation, Grunert, Nordli and Solem established convergence of a method that combines piecewise-linear projection with evolution along characteristics [6]. These studies connect preservation of analytical structure with the choice of variables, boundary treatment and approximation of the evolving solution.

Mesh allocation introduces a further choice. In adaptive moving-mesh methods, a monitor function determines how a fixed number of nodes is distributed, and equidistribution provides a systematic rule for that allocation [7]. A reciprocal transformation supplies such a monitor through a conserved density and determines the associated node velocity through its flux. For the smooth 2HS waves studied here, however, the physical density decreases near a velocity peak. Equal density mass then assigns wider cells to this region. This observation motivates examining how the choice of conserved density affects the accuracy of the two fields. A density involving curvature or the quadratic balance of slope and physical density can assign resolution to different parts of the same wave. The accompanying flux also changes the node motion, so its numerical effect involves both the initial allocation and the subsequent coupled evolution.

The availability of exact solutions at both the continuous and semi-discrete levels makes a more detailed comparison possible. A numerical trajectory may follow its semi-discrete exact solution very accurately while retaining an appreciable difference from the continuous wave. On a moving mesh, this comparison also involves the physical locations at which the fields are evaluated. The distinction is especially relevant for the density, which represents a conserved mass per cell and therefore has a natural interpretation as a cell average.

We seek to understand how these analytical structures can guide the design and interpretation of numerical computations. The exact lattice solutions allow the error of a time integrator to be measured independently of the approximation of the continuous wave. The reciprocal transformation identifies the physical locations and cell averages at which this latter comparison should be made. The density conservation law, in turn, specifies how nodes move and explains where the natural lattice places its resolution. Together, these relations give a basis for choosing numerical tests whose results can be traced back to the equations.

Our analysis begins with an explicit identity between the integrable and direct moving-grid vector fields. It exposes a density-dependent first-order correction, which helps explain the different refinement rates observed in the soliton experiments. This connects an identifiable term in the spatial model with its numerical accuracy. We then make the choice of conserved coordinate part of the numerical investigation. Starting from the field equations, we determine a curvature–mass ratio by balancing stretching rates and an energy-type density by cancelling the exchange in a quadratic slope–density balance. Their local fluxes, together with physical mass, define three moving meshes. A matched comparison with uniform fixed FD examines how these allocations affect the two nodal fields. These meshes retain a common second-order refinement trend while producing different errors in velocity and density. The curvature–mass ratio gives the most balanced results in the two tests, and the energy-type choice improves the single-wave density error further at a cost in velocity accuracy. This identifies the local choice of conserved coordinate as a way to influence the error coefficients of the coupled computation. A final set of perturbation experiments follows the transfer of a velocity disturbance into the density field within each method.

Section 2 introduces the continuous equations, reciprocal coordinate and exact solutions used in the experiments. Section 3 presents the three spatial formulations, their boundary data and time integrators. Section 4 examines spatial errors, time integration and perturbation responses. Section 5 develops three conservation-law-based self-adaptive meshes and examines their fieldwise accuracy. Section 6 discusses the numerical implications of these results. Appendices A–C provide the correction calculation, nonuniform difference formulas and supplementary numerical checks.

2. The 2HS system and exact wave solutions

We use a normalization in which the constant background is $u = 0$, $\rho = 1$, and the parameter c is positive. The formulas below specify the branch and coordinate conventions used throughout the computation. The integrable semi-discretization and its soliton family are taken from [3]; the numerical comparisons concern their approximation properties.

2.1. Continuous equations and reciprocal coordinate

The integrated form of the system is

$$u_{tx} - uu_{xx} - \tfrac{1}{2}u_x^2 - 4u - \tfrac{c^2}{2}(\rho^2 - 1) = 0, \qquad \rho_t = (u\rho)_x. \tag{1}$$

The integration constant in the first equation is fixed by the stated background. For positive density, the conservation law defines a reciprocal coordinate through

$$dX = \rho\, dx + u\rho\, dt, \qquad dT = dt, \qquad x_T = -u, \quad x_X = \rho^{-1}. \tag{2}$$

Thus a node at fixed X moves with velocity $-u$. Writing $w = u_x$ and differentiating along this motion gives

$$w_T = \tfrac{1}{2}w^2 + 4u + \tfrac{c^2}{2}(\rho^2 - 1), \qquad \rho_T = \rho w. \tag{3}$$

These equations connect the field evolution to the changing mesh. In particular, the mass of an interval between two fixed reciprocal labels is constant. Uniform spacing in X therefore means equal density mass in each moving physical cell.

2.2. Continuous and finite-lattice solitons

For the one- and two-soliton tests, a normalized positive tau function gives a compact common representation. Set $q_i = p_i/(cp_i - 1)$, $\omega_i = p_i^{-1} - q_i^{-1}$, $\beta_i = p_i - q_i$, and let $\theta_i = \beta_i X + \omega_i T + \theta_{i0}$. The continuous tau functions are

$$f = 1 + e^{\theta_1} \quad (N = 1), \qquad f = 1 + e^{\theta_1} + e^{\theta_2} + A_{12}e^{\theta_1+\theta_2} \quad (N = 2), \tag{4}$$

$$A_{12} = \frac{(p_2 - p_1)(q_2 - q_1)}{(q_2 - p_1)(p_2 - q_1)}, \qquad u = (\log f)_{TT}, \quad x = X - (\log f)_T, \quad \rho = \left[1 - (\log f)_{XT}\right]^{-1}. \tag{5}$$

The selected parameters give positive tau functions and monotone physical-coordinate maps on the evaluation domains. To evaluate a reference at a prescribed physical point x, we invert the map in (5) and then evaluate the fields at the recovered label X.

For the finite lattice, let $X_k = ka$ and replace $\beta_i X$ by $k\sigma_i(a)$, where

$$\sigma_i(a) = \log\frac{1 - aq_i}{1 - ap_i}, \qquad a \max_i(p_i, q_i) < 1. \tag{6}$$

Denoting the resulting tau function by f_k, the exact lattice fields are

$$u_k^* = (\log f_k)_{TT}, \qquad x_k^* = ka - (\log f_k)_T, \qquad d_k^* = x_k^* - x_{k-1}^*, \qquad \rho_k^* = a/d_k^*. \tag{7}$$

The continuous and discrete solutions share their time dependence and interaction coefficient. Their spatial phases differ through $\sigma_i(a)$. Since $\sigma_i(a)/a = \beta_i + \frac{a}{2}(p_i^2 - q_i^2) + O(a^2)$, this representation already exposes a first-order change in the wave profile at finite lattice spacing.

3. Numerical formulation

We formulate the numerical evolution so that the analytical relations between fields and coordinates remain explicit. On a moving mesh, velocity jumps and cell widths determine both the field reconstruction and the motion of the physical nodes. On a fixed grid, the differentiated velocity variable $m = u_{xx} + 2$ evolves with the density, and a discrete boundary-value problem recovers the velocity. These choices turn the wave equations into finite-dimensional evolutions while retaining the reconstruction needed to compare their physical fields.

The two moving formulations provide a particularly useful comparison because they share the same reconstruction and node velocity. Their slope equations can be compared at an identical state, exposing the finite-spacing correction introduced by the integrable lattice. Fixed FD supplies a centered approximation in physical coordinates. We specify each evolution and its boundary data before applying common time integrators, so that the experiments can distinguish the spatial formulation from its temporal approximation.

3.1. Integrable and direct moving-grid formulations

In the reciprocal coordinate, each cell carries the prescribed mass a. Its width therefore determines its mean density, and the velocity difference between its endpoints determines the rate at which that width changes. This suggests using the two cell quantities as the evolving variables. Define $v_k = u_k - u_{k-1}$, $d_k = x_k - x_{k-1}$, $\rho_k = a/d_k$ and $\alpha = a(a - c)$. In these variables, the integrable semi-discretization of [3] takes the form

$$\dot{v}_k = 2d_k(u_k + u_{k-1}) + \frac{(d_k^2 - \alpha)^2 - v_k^2}{2d_k} - \frac{c^2 d_k}{2}, \qquad \dot{d}_k = -v_k. \tag{8}$$

With the prescribed left value $u_0 = b(T)$, the fields and physical nodes are reconstructed as $u_k = b + \sum_{j=1}^{k} v_j$ and $x_k = x_0 + \sum_{j=1}^{k} d_j$, with $\dot{x}_0 = -b$. Consequently, every node satisfies $\dot{x}_k = -u_k$. We call this formulation integrable SD.

For comparison, a direct approximation of (3) uses the edge slope $w_k = v_k/d_k$ and the adjacent-node mean for u. This moving FD formulation is

$$\dot{w}_k = \tfrac{1}{2}w_k^2 + 2(u_k + u_{k-1}) + \tfrac{c^2}{2}(\rho_k^2 - 1), \qquad \dot{d}_k = -d_k w_k, \quad \rho_k = a/d_k. \tag{9}$$

It is implemented in the same jump variables by $\dot{v}_k = d_k\dot{w}_k - w_k v_k$. The two moving methods therefore share the field reconstruction and mesh velocity. Their difference lies in the slope equation. At the same state, direct substitution into (8) gives the exact identity

$$F^{(w)}_{\text{SD}} - F^{(w)}_{\text{moving FD}} = ac(1 - \rho_k^2) + \tfrac{1}{2}a^2(\rho_k^{-1} - \rho_k)^2. \tag{10}$$

The first term is linear in the reciprocal spacing and depends on the density departure from its background. It vanishes at the unit-density background and contributes inside a nonuniform wave. Equation (10) therefore identifies where the leading correction enters the evolution and how it scales with reciprocal spacing. The continuous-reference experiments examine the resulting field errors, including their accumulation through the reconstruction. The algebra leading to (10) and the corresponding phase expansion are detailed in Appendix A.

3.2. Fixed-grid finite differences

A fixed physical grid gives a complementary representation in which the positions are prescribed throughout the computation. We introduce $m = u_{xx} + 2$ to obtain a local evolution equation coupled to the transported density. Differentiating the first equation in (1) yields

$$m_t = um_x + 2u_x m + c^2 \rho \rho_x, \qquad \rho_t = (u\rho)_x, \qquad u_{xx} = m - 2. \tag{11}$$

All spatial derivatives are approximated by centered differences, $\mathcal{D}_0 z_j = (z_{j+1} - z_{j-1})/(2\Delta x)$ and $\mathcal{D}_2 z_j = (z_{j+1} - 2z_j + z_{j-1})/\Delta x^2$. At each time stage, the interior values of u are obtained from the tridiagonal system $\mathcal{D}_2 u = m - 2$. Its reconstructed values are then used to evaluate the right-hand sides of (11). We refer to this formulation as fixed FD. The boundary-value recovery couples the velocity across the interval, while the right-hand sides advance the local values of curvature and density. This distinction is useful when interpreting the different error profiles of the two physical fields.

3.3. Initial data, boundary data and time integration

The choice of initial and boundary data follows the reference solution used in each experiment. To measure approximation of the continuous wave, we initialize its physical fields and supply its boundary values throughout the evolution. To measure time-integration error, we use an exact trajectory of the finite lattice. This second choice fixes the spatial model while the time step is varied.

Spatial comparisons begin with the continuous exact fields at $T = 0$. For the moving methods, these are sampled at uniform reciprocal labels and converted into jumps and physical cell widths. The left value is $b(T) = u(X_L, T)$, and the left node evolves with $-b(T)$. The fixed-grid method uses exact physical boundary values at $x = -4, 4$; its boundary values of m are formed with the same centered second difference applied to the exact field, including one exterior reference point on either side. Thus the fixed-grid experiment uses two-sided boundary data, whereas the moving experiments use a left-driven open chain.

Temporal comparisons instead start from the exact lattice state (7), with its matching lattice boundary. In both settings, write the evolving state as z and its reconstructed right-hand side as $\dot{z} = H(T, z)$. We use forward Euler, classical fourth-order Runge–Kutta and implicit trapezoidal (Crank–Nicolson) integration:

$$z^{n+1} = z^n + \Delta T H(T_n, z^n), \qquad z^{n+1} = z^n + \tfrac{\Delta T}{2}\big[H(T_n, z^n) + H(T_{n+1}, z^{n+1})\big]. \tag{12}$$

$$K_1 = H(T_n, z^n), \quad K_2 = H(T_n + \tfrac{\Delta T}{2}, z^n + \tfrac{\Delta T}{2} K_1), \quad K_3 = H(T_n + \tfrac{\Delta T}{2}, z^n + \tfrac{\Delta T}{2} K_2), \tag{13}$$

$$K_4 = H(T_n + \Delta T, z^n + \Delta T K_3), \qquad z^{n+1} = z^n + \tfrac{\Delta T}{6}(K_1 + 2K_2 + 2K_3 + K_4). \tag{14}$$

Boundary data and reconstructed fields are evaluated at each stage time. The implicit iteration uses a scaled tolerance of 2×10^{-13}. The three methods have global orders one, four and two, respectively, for a smooth finite-dimensional evolution on a fixed interval with the usual bounded-derivative assumptions. Their different orders allow the exact lattice trajectory to serve as a temporal refinement test. In the continuous-reference experiments, the time step is instead chosen small enough that halving it has little effect on the measured spatial errors.

4. Numerical experiments

The exact wave families provide two references with different uses. Continuous solutions measure how well a spatial formulation resolves the physical wave, whereas exact finite-lattice solutions measure how accurately a time integrator follows that formulation. We use both references to connect the algebraic correction in Section 3.1 with observable errors, and then perturb the initial velocity to examine the response generated in the density field. Section 4.1 specifies the profiles and physical error measures. Section 4.2 compares the spatial models and their continuum approximation, and Section 4.3 isolates the effect of time integration on the integrable lattice. Section 4.4 follows the response to a prescribed initial velocity perturbation within each spatial method.

4.1. Test problems and error measures

The one-soliton profile provides a single localized structure whose position, width and density depression can be resolved together. The two-soliton profile introduces two regions with different wave heights, testing how the same numerical choices distribute their errors over a more varied solution. Both belong to the exact family in Section 2.2, so their field values and physical coordinates can be evaluated throughout the observation interval.

We set $c = 1$, take zero phase constants and zero coordinate shift, and use a one-soliton case with $p_1 = 5$ and a two-soliton case with $(p_1, p_2) = (1.1, 1.25)$. The reciprocal interval is $[-4, 4]$, and the observation interval is $0 \le T \le 0.25$. Terminal error tables use $T = 0.25$, while the spatial profile figures use the midpoint. Figures 1 and 2 show the exact physical fields at the midpoint of the observation interval, $T = 0.125$. The selected interval follows a short portion of the two-soliton trajectory and permits the spatial and temporal refinements to use the same observation times.

One-soliton case

Figure 1 shows the localized velocity peak and its associated density depression. The two fields occupy the same wave region but have different shapes, so their errors are recorded separately. The density depression also determines where the natural equal-mass mesh has its wider cells.

The velocity consists of one smooth localized peak, accompanied by a density depression. These two features locate the region where the field curvature and natural cell width change most strongly.

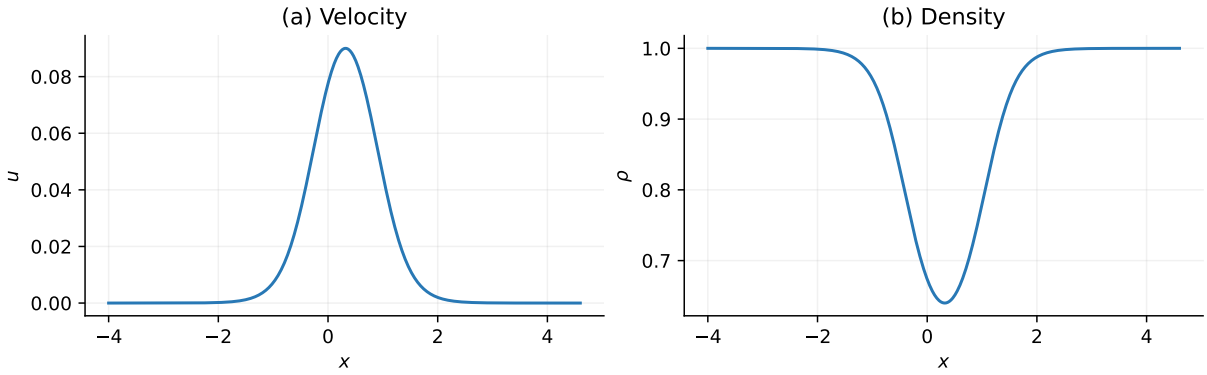


Figure 1. One-soliton case: continuous exact velocity and density at T=0.125.

Two-soliton case

The two-soliton profile contains two resolved peaks with different heights and a corresponding pair of density depressions. This gives the mesh two distinct regions to resolve within the same physical interval.

For a moving grid, the velocity is compared with the continuous exact value at each numerical node. The density reference is its continuous mean over the numerical cell, obtained from the inverse reciprocal map:

$$E_u = \max_k |u_k - u^*(x_k, T)|, \qquad E_\rho = \max_k \left| \rho_k - \frac{X(x_k, T) - X(x_{k-1}, T)}{x_k - x_{k-1}} \right|. \tag{15}$$

For fixed FD, both errors are nodal maximum errors against continuous point values. Temporal errors use the corresponding exact finite-lattice fields at the same label indices. These definitions retain the natural location of each variable. In the field plots, moving-cell density values are displayed at cell centers; error curves use the cell-average reference in (15).

For the integrable lattice, the velocity error also admits a useful signed decomposition,

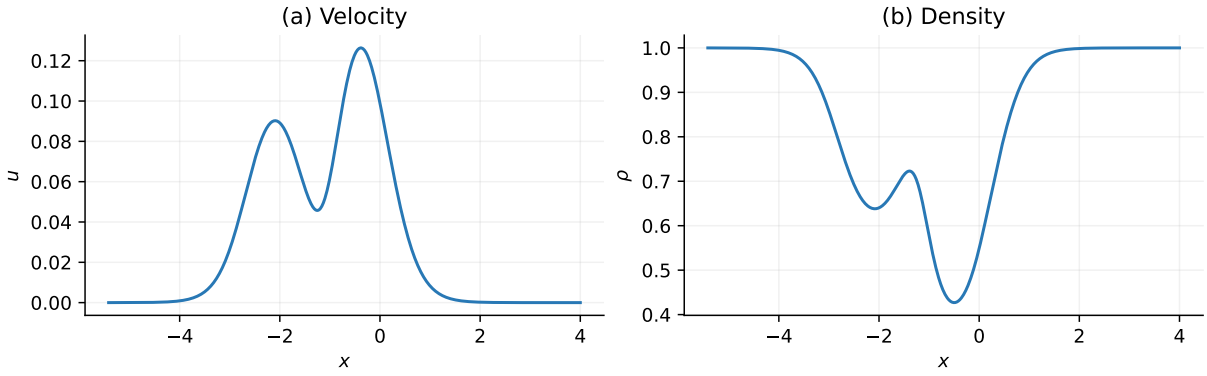


Figure 2. Two-soliton case: continuous exact velocity and density at T=0.125.

$$u_{\text{num}} - u^*(x_{\text{num}}, T) = (u_{\text{num}} - u_a^*) + [u_a^* - u^*(x_a^*, T)] + [u^*(x_a^*, T) - u^*(x_{\text{num}}, T)]. \tag{16}$$

The three terms represent integration error, finite-lattice model error and coordinate displacement. The analogous density identity uses cell averages over the corresponding intervals. The signed terms also distinguish changes in field value from changes in the sampling position. In the spatial comparison we retain their combined effect against the continuous wave. In the temporal comparison the exact lattice state supplies the reference, so the measured discrepancy records the numerical advancement of that state.

4.2. Spatial formulations and the continuous reference

We compare the integrable semi-discretization with centered finite differences on a fixed physical grid, denoted here by integrable SD and FD. Both calculations begin with the continuous exact initial fields and use RK4 with $\Delta T = 5 \times 10^{-4}$. Refining the spatial spacing tests how each formulation approaches the continuous wave. For integrable SD, the phase expansion in (6) and the slope correction in (10) both identify a first-order finite-spacing contribution. The centered fixed-grid differences provide a second-order comparison. We examine whether these distinctions remain visible after the fields and coordinates have evolved, and where the largest discrepancies occur within each profile.

Tables 1 and 2 report terminal errors at $T = 0.25$ for three resolutions. The SD reciprocal spacing a and the FD physical spacing Δx take the same nominal values, giving the same node count at each resolution. SD density errors use exact cell averages, whereas FD density errors use exact nodal values, as specified in (15). Figures 3 and 4 show the corresponding midpoint fields and errors at $T = 0.125$. Halving the time step at spacing 0.02 changes the evolved states by less than 3×10^{-14}, placing these spatial errors well above the observed temporal refinement difference.

One-soliton case

We first examine the single localized wave. The terminal table measures the refinement rates, while the midpoint curves show where the differences develop in physical space.

Figure 3 compares the numerical velocity and density with the exact profiles at the midpoint. The lower panels display the corresponding absolute errors, locating the differences that are summarized by the terminal table.

The FD samples closely follow the exact wave at this resolution. Integrable SD develops a broader velocity discrepancy on the downstream side and larger density errors than the FD formulation. Its density discrepancy is nevertheless much smaller in absolute magnitude than its velocity discrepancy. Table 1 confirms that the FD formulation retains smaller errors in their respective field metrics across the tested resolutions.

Table 1. Continuous-reference errors after RK4 evolution to T=0.25. Density errors are cell-average errors for moving methods and nodal errors for fixed FD. Bold values indicate the smaller field error at each nominal spacing.

Case	Spatial method	Spacing	E_u	Order	E_ρ	Order
One soliton	Integrable SD	$a = 0.04$	5.831×10^{-2}	—	1.310×10^{-3}	—
		$a = 0.02$	2.896×10^{-2}	1.01	6.507×10^{-4}	1.01
		$a = 0.01$	1.443×10^{-2}	1.00	3.241×10^{-4}	1.01
	FD	$\Delta x = 0.04$	$\mathbf{2.531 \times 10^{-6}}$	—	$\mathbf{2.313 \times 10^{-5}}$	—
		$\Delta x = 0.02$	$\mathbf{6.335 \times 10^{-7}}$	2.00	$\mathbf{5.788 \times 10^{-6}}$	2.00
		$\Delta x = 0.01$	$\mathbf{1.584 \times 10^{-7}}$	2.00	$\mathbf{1.448 \times 10^{-6}}$	2.00

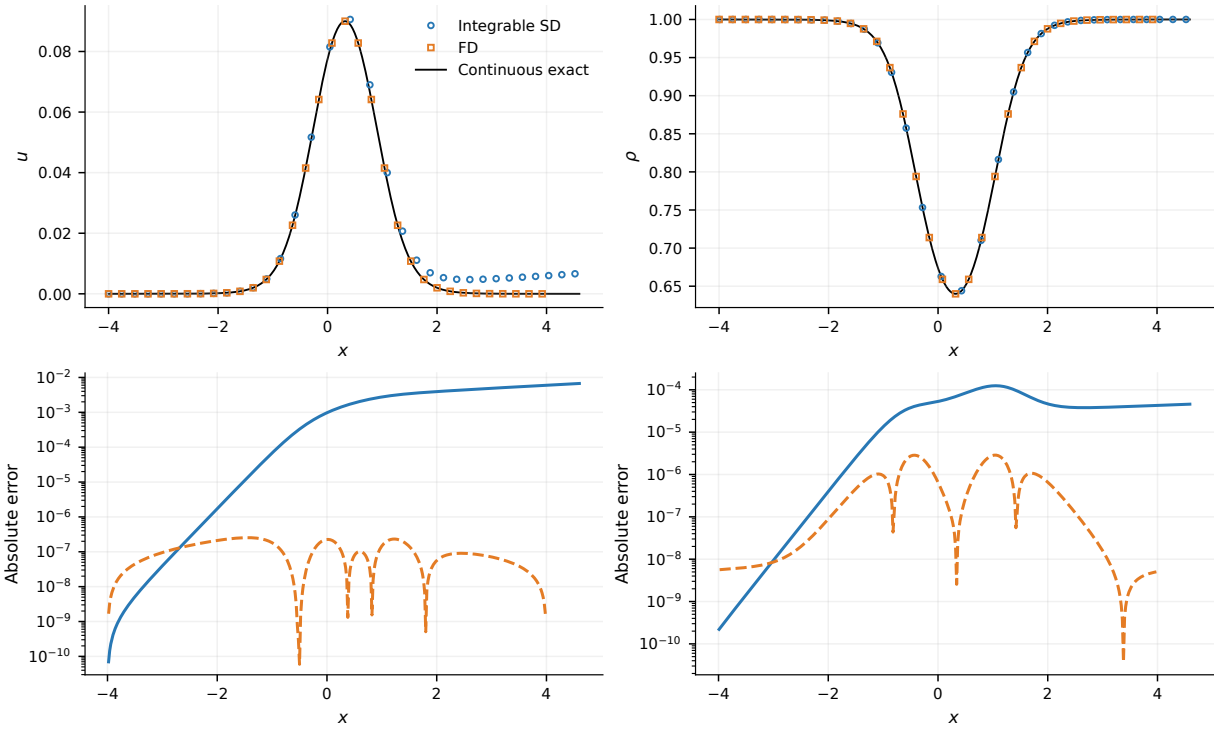


Figure 3. One-soliton case: fields and spatial errors at T=0.125, with nominal spacing 0.02 and RK4 step 0.0005. Black curves are continuous point values; colored symbols are computed samples. Moving-cell density errors use exact cell averages. Prescribed boundary values are excluded from the logarithmic error curves.

Two-soliton case

The second case places two wave structures in the computational interval. We use the same refinement and midpoint sampling to examine how their different amplitudes affect the two field errors.

Table 2. Continuous-reference errors after RK4 evolution to T=0.25. Density errors are cell-average errors for moving methods and nodal errors for fixed FD. Bold values indicate the smaller field error at each nominal spacing.

Case	Spatial method	Spacing	E_u	Order	E_ρ	Order
Two solitons	Integrable SD	$a = 0.04$	1.727×10^{-1}	—	3.779×10^{-3}	—
		$a = 0.02$	8.522×10^{-2}	1.02	1.862×10^{-3}	1.02
		$a = 0.01$	4.232×10^{-2}	1.01	9.240×10^{-4}	1.01
	FD	$\Delta x = 0.04$	$\mathbf{1.418 \times 10^{-5}}$	—	$\mathbf{6.763 \times 10^{-5}}$	—
		$\Delta x = 0.02$	$\mathbf{3.582 \times 10^{-6}}$	1.99	$\mathbf{1.711 \times 10^{-5}}$	1.98
		$\Delta x = 0.01$	$\mathbf{8.962 \times 10^{-7}}$	2.00	$\mathbf{4.283 \times 10^{-6}}$	2.00

Figure 4 makes the same midpoint comparison for the two-soliton case. Its field panels show how the two wave structures are resolved, and its error panels show where the numerical profiles depart from the reference.

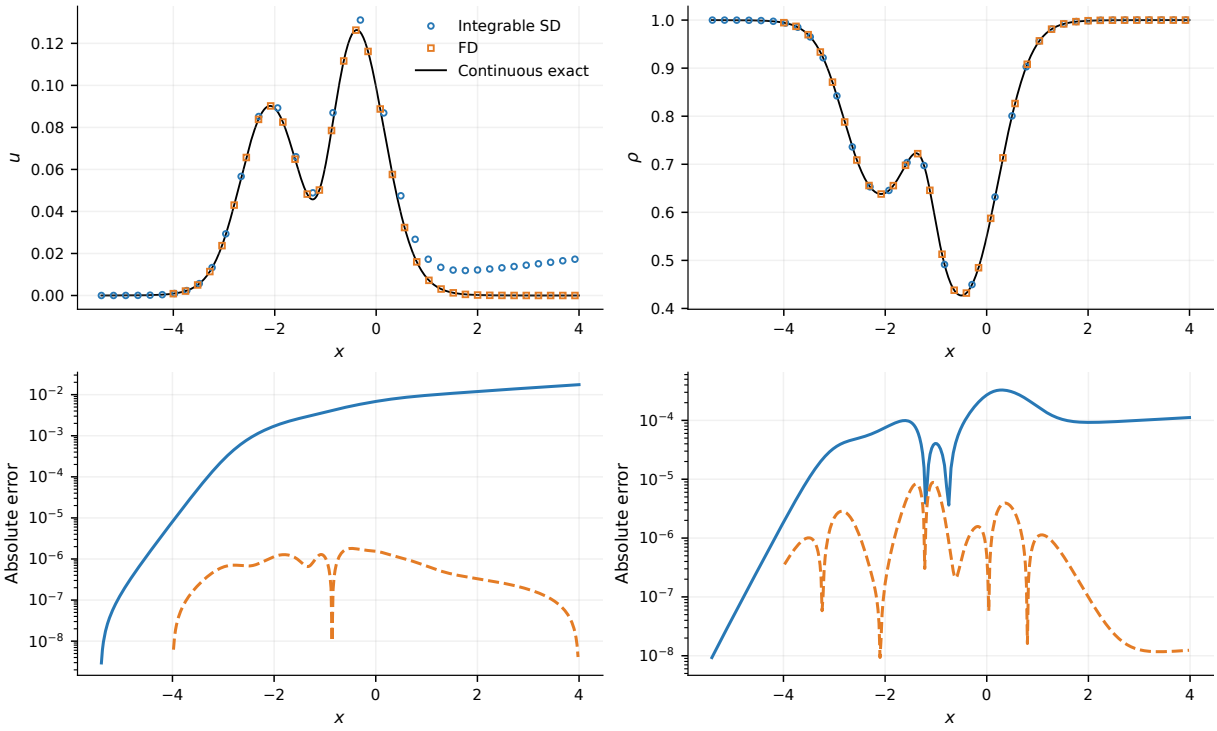


Figure 4. Two-soliton case: fields and spatial errors at T=0.125, with nominal spacing 0.02 and RK4 step 0.0005. Black curves are continuous point values; colored symbols are computed samples. Moving-cell density errors use exact cell averages. Prescribed boundary values are excluded from the logarithmic error curves.

Integrable SD again has the larger field errors, with its velocity discrepancy extending downstream of the localized waves. The density errors vary more locally around the two depressions. The FD formulation resolves both structures more closely in the displayed comparison, and Table 2 records smaller terminal errors throughout the tested resolutions. Thus the additional wave structure changes the magnitude and distribution of the errors while preserving the overall pattern seen in the one-soliton case.

The two soliton cases show the same overall pattern: FD produces smaller errors in the reported field

metrics, and its advantage increases under spatial refinement. The observed orders explain this trend. Integrable SD gives approximately first-order error reduction, while FD gives approximately second-order reduction. Halving the spacing therefore reduces the SD errors by about two and the FD errors by about four. The error coefficients and wave profile determine the absolute magnitudes within this pattern.

This difference in order is consistent with the analytical structure of the spatial approximations. The exact lattice phase expansion following (7) has a leading correction proportional to a, and the SD slope equation contains the density-dependent first-order correction identified in (10). The centered differences used by FD have second-order local truncation errors for smooth fields. These properties help explain the refinement rates measured in Tables 1 and 2. Within SD, the velocity is reconstructed by accumulating jumps from the left boundary, so changes in the slope evolution can contribute to discrepancies farther along the chain. This is consistent with the downstream velocity errors seen in both figures, while the density responds through the changing cell widths.

The comparison thus relates the field errors to the approximation order and reconstruction of each spatial formulation. For the smooth profiles and resolutions examined here, the second-order FD approximation resolves the continuous fields more accurately than the first-order integrable lattice. The fixed and moving FD mesh choices are examined separately in Section 5.

4.3. Time integration on the integrable lattice

The spatial tests measure the combined discrepancy from the continuous fields. The exact lattice solution now allows us to hold that spatial model fixed and examine only its numerical advancement. To isolate temporal accuracy, we fix $a = 0.02$ and initialize the exact lattice solution. Euler, RK4 and Crank–Nicolson now approximate the same trajectory with matching boundary data. Tables 3 and 4 report errors against that trajectory, so the refinement measures the effect of time integration without including the continuous-to-lattice profile difference.

One-soliton case

For the single wave, all three time integrators start on the same exact lattice trajectory. Their differences therefore measure how accurately they advance that trajectory.

Table 3. Temporal errors at T=0.25 for a=0.02. Bold entries are smallest among the three integrators at the same case and step size.

Case	Integrator	ΔT	E_u	Order	E_ρ	Order
One soliton	Euler	0.025	5.587×10^{-5}	—	1.319×10^{-4}	—
		0.0125	2.813×10^{-5}	0.99	6.591×10^{-5}	1.00
		0.00625	1.412×10^{-5}	0.99	3.294×10^{-5}	1.00
	RK4	0.025	$\mathbf{7.719 \times 10^{-9}}$	—	$\mathbf{4.991 \times 10^{-10}}$	—
		0.0125	$\mathbf{4.982 \times 10^{-10}}$	3.95	$\mathbf{3.307 \times 10^{-11}}$	3.92
		0.00625	$\mathbf{3.164 \times 10^{-11}}$	3.98	$\mathbf{2.145 \times 10^{-12}}$	3.95
	Crank–Nicolson	0.025	1.387×10^{-7}	—	3.872×10^{-7}	—
		0.0125	3.467×10^{-8}	2.00	9.681×10^{-8}	2.00
		0.00625	8.667×10^{-9}	2.00	2.420×10^{-8}	2.00

The curves summarize the largest error over the lattice at each time. Euler accumulates the largest error, Crank–Nicolson reduces it substantially, and RK4 follows the exact trajectory most closely at the displayed step. The refinement table identifies the corresponding temporal orders.

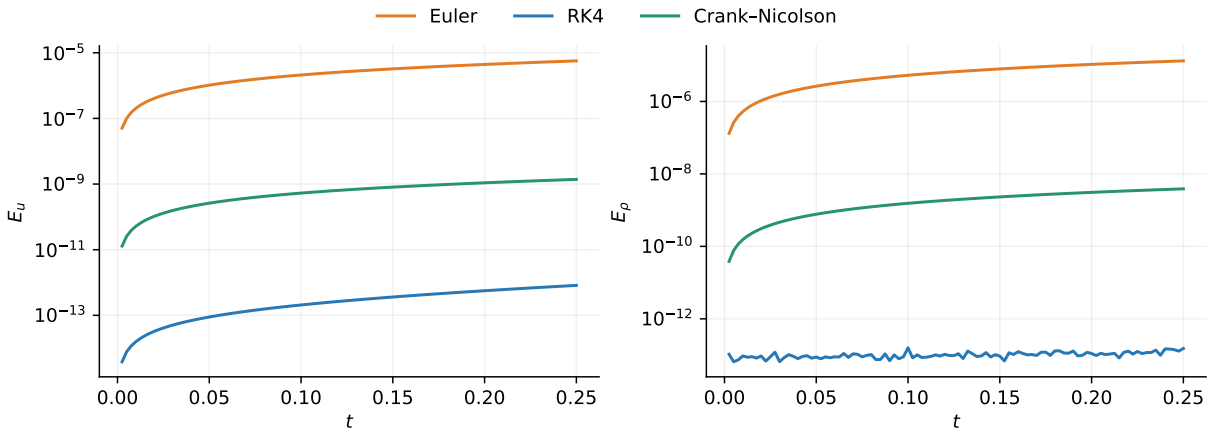


Figure 5. One-soliton case: maximum temporal errors on the integrable lattice; a=0.02 and ΔT=0.0025.

Two-soliton case

The two-wave profile tests the same time algorithms on a more structured lattice trajectory. Both components are measured against the matching exact finite-lattice solution.

Table 4. Temporal errors at T=0.25 for a=0.02. Bold entries are smallest among the three integrators at the same case and step size.

Case	Integrator	ΔT	E_u	Order	E_ρ	Order
Two solitons	Euler	0.025	8.926×10^{-5}	—	1.959×10^{-4}	—
		0.0125	4.535×10^{-5}	0.98	9.767×10^{-5}	1.00
		0.00625	2.287×10^{-5}	0.99	4.876×10^{-5}	1.00
	RK4	0.025	$\mathbf{2.414 \times 10^{-8}}$	—	$\mathbf{1.231 \times 10^{-9}}$	—
		0.0125	$\mathbf{1.567 \times 10^{-9}}$	3.95	$\mathbf{8.195 \times 10^{-11}}$	3.91
		0.00625	$\mathbf{9.974 \times 10^{-11}}$	3.97	$\mathbf{5.279 \times 10^{-12}}$	3.96
	Crank–Nicolson	0.025	3.405×10^{-7}	—	5.507×10^{-7}	—
		0.0125	8.511×10^{-8}	2.00	1.377×10^{-7}	2.00
		0.00625	2.128×10^{-8}	2.00	3.442×10^{-8}	2.00

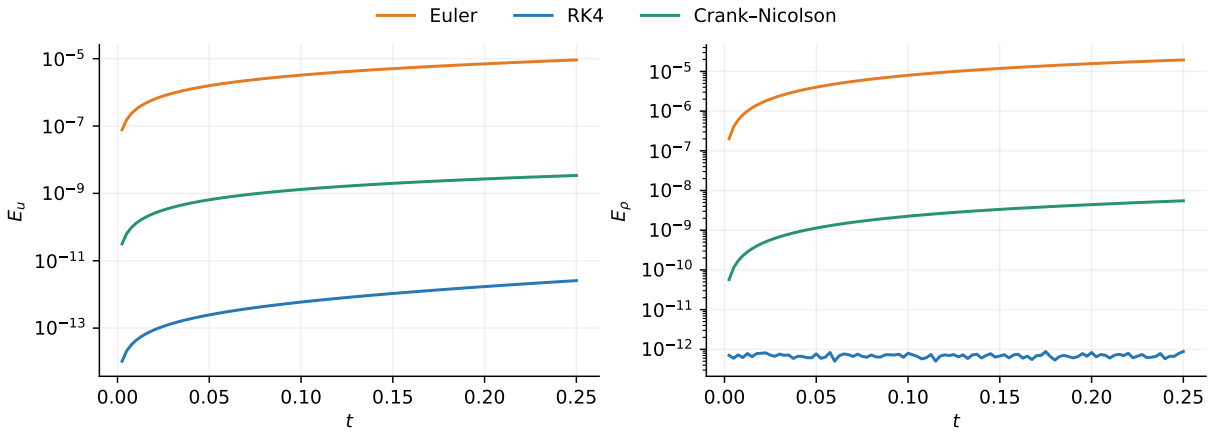


Figure 6. Two-soliton case: maximum temporal errors on the integrable lattice; a=0.02 and ΔT=0.0025.

The curves summarize the largest error over the lattice at each time. Euler accumulates the largest error,

Crank–Nicolson reduces it substantially, and RK4 follows the exact trajectory most closely at the displayed step. The refinement table identifies the corresponding temporal orders.

The time histories use $\Delta T = 0.0025$ and show the errors in both components throughout the interval. Their magnitudes describe how accurately each integrator follows the coupled field and mesh evolution. The continuous-reference errors in Section 4.2 answer a different numerical question: once the temporal error has become small, the spatial model and field reconstruction determine the remaining discrepancy from the continuous wave.

Tables 3 and 4 also show how the exact lattice solution guides the choice of temporal resolution. Euler approaches first-order refinement, Crank–Nicolson gives approximately second-order refinement, and RK4 approaches fourth order over the tested steps. At the smaller step used for the histories, the RK4 density error reaches a nearly flat level around floating-point evaluation accuracy. For the spatial experiments, the time-step-halving check establishes that further temporal refinement has little effect on the reported errors. Computational effort can then be directed toward the spatial approximation and the distribution of nodes.

4.4. Propagation of an initial velocity perturbation

Exact-wave comparisons examine the evolution of a prescribed background. To study the response to a nearby initial state, we add a smooth velocity perturbation while preserving the initial physical nodes and density. This makes it possible to follow the velocity response and the density response generated by the coupling. For each method, the perturbed solution is compared with its own unperturbed numerical trajectory. Subtracting the two trajectories follows the added disturbance on that numerical background. Because the initial density is unchanged, its subsequent response measures the transfer generated by the coupled evolution; on the moving grids, that transfer also involves the changing cell widths.

Let $\eta = (x - x_L)/(x_R - x_L)$ on the initial physical interval and define $\phi(x) = \sin^2(\pi\eta)\cos(2\pi\eta)$, normalized to unit maximum over the initial nodes. We set $u^\varepsilon(x,0) = u^0(x,0) + \varepsilon\phi(x)$, $\rho^\varepsilon(x,0) = \rho^0(x,0)$, with $\varepsilon = 10^{-6}$. The moving state is perturbed through the corresponding velocity jumps. For fixed FD, the perturbation is applied as $m^\varepsilon = m^0 + \varepsilon\mathcal{D}_2\phi$. All boundary data retain their background values. The normalized responses are

$$R_u(T) = \varepsilon^{-1} \max_k |u_k^\varepsilon(T) - u_k^0(T)|, \qquad R_\rho(T) = \varepsilon^{-1} \max_k |\rho_k^\varepsilon(T) - \rho_k^0(T)|. \tag{17}$$

Moving-grid differences in (17) are taken at matching labels, so the response includes the coupled change of the field and its trajectory. Fixed-grid differences are taken at matching physical nodes. Figures 7 and 8 present these histories with nominal spacing 0.02 and RK4 time step 5×10^{-4}.

One-soliton case

Table 5. Terminal normalized responses and maximum changes in the complete normalized response histories under amplitude or time-step halving.

Case	Method	$R_u(T)$	$R_\rho(T)$	ε-halving change	ΔT-halving change
One soliton	Integrable SD	11.39182	0.23983	1.272×10^{-6}	3.220×10^{-9}
	Moving FD	11.40663	0.23977	1.273×10^{-6}	5.662×10^{-9}
	Fixed FD	0.75066	0.17379	1.155×10^{-8}	5.028×10^{-9}

The moving-label velocity response grows over the interval and generates a density response. The fixed-position calculation also develops a density response while its velocity response remains below its

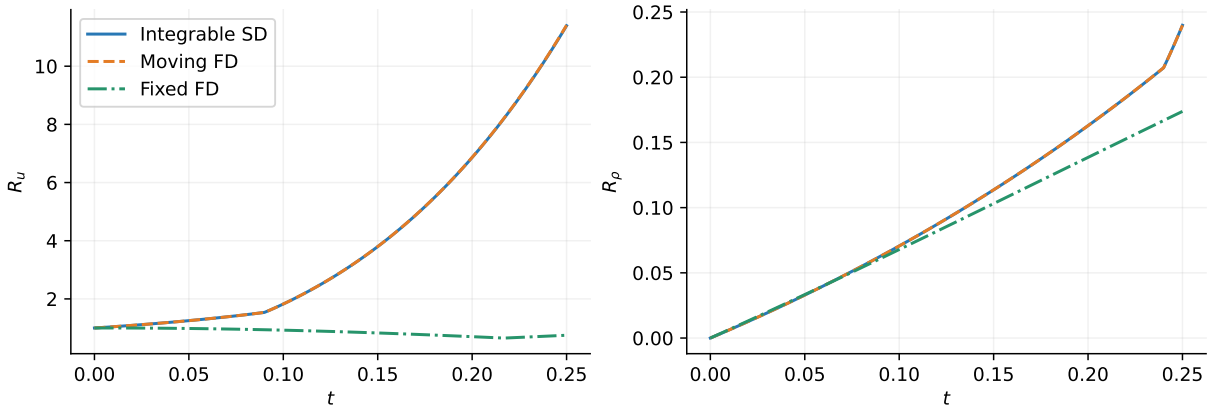


Figure 7. One-soliton case: normalized velocity and density responses to a velocity perturbation of amplitude 10^{-6}, at matching native indices.

initial maximum at the terminal time. The table reports these histories' terminal values and their sensitivity to amplitude and time-step halving within each formulation.

Two-soliton case

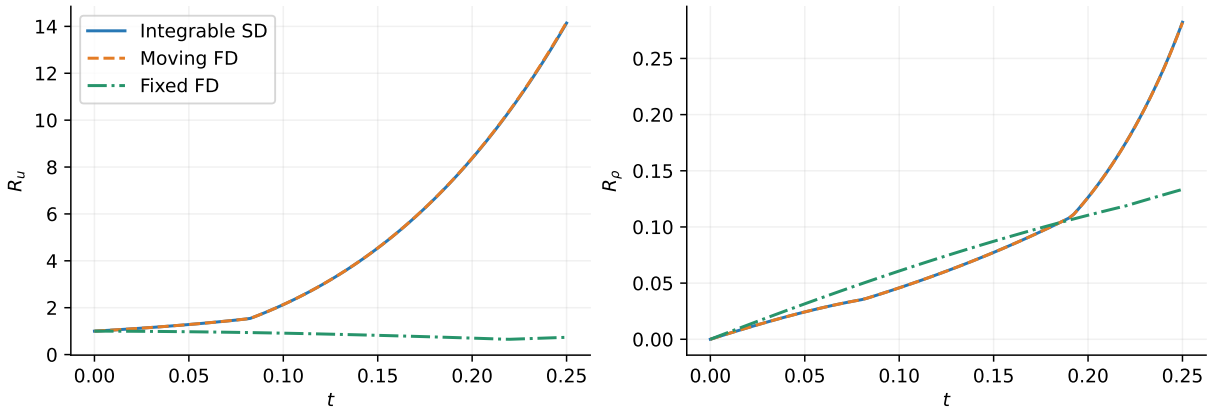


Figure 8. Two-soliton case: normalized velocity and density responses to a velocity perturbation of amplitude 10^{-6}, at matching native indices.

Table 6. Terminal normalized responses and maximum changes in the complete normalized response histories under amplitude or time-step halving.

Case	Method	$R_u(T)$	$R_\rho(T)$	ε-halving change	ΔT-halving change
Two solitons	Integrable SD	14.14049	0.28224	1.692×10^{-6}	7.327×10^{-9}
	Moving FD	14.17654	0.28188	1.693×10^{-6}	4.108×10^{-9}
	Fixed FD	0.73678	0.13345	9.541×10^{-9}	4.358×10^{-9}

The moving-label velocity response grows over the interval and generates a density response. The fixed-position calculation also develops a density response while its velocity response remains below its initial maximum at the terminal time. The table reports these histories' terminal values and their sensitivity to amplitude and time-step halving within each formulation.

Initially, the perturbation is confined to the velocity. The subsequent density response records the cou-

pling through the velocity gradient and, on a moving mesh, through the changing cell widths. The curves describe this transfer within each numerical formulation over the stated interval. Their amplitude and time-step checks quantify the reproducibility of the observed response.

5. Self-adaptive moving meshes from conservation laws

The physical density plays two roles in the 2HS computation. It is one of the evolving fields, and its conservation law supplies the reciprocal coordinate used by the integrable lattice. These roles need not favor the same spatial resolution. For the smooth waves considered here, the density decreases near a velocity peak, so a grid carrying equal physical mass has wider cells in that region. This prompts us to ask how other conserved quantities of the same equations distribute the nodes, and how their distributions affect the errors in velocity and density.

We develop three self-adaptive moving-mesh (SAMM) choices directly from the continuous equations. Physical mass provides the starting point. A curvature–mass ratio follows by balancing two stretching rates, and an energy-type quadratic density follows by cancelling the exchange terms between the velocity gradient and physical density. We explain these choices before comparing their numerical evolutions. The comparison holds the field discretization and time integrator fixed, allowing each conservation law to be assessed through the allocation and motion that it supplies.

5.1. Finding conserved densities from the field equations

For convenience, we recall the continuous system and its background,

$$u_{tx} = uu_{xx} + \frac{1}{2}u_x^2 + 4u + \frac{c^2}{2}(\rho^2 - 1), \qquad \rho_t = (u\rho)_x, \qquad (u, \rho) \to (0, 1).$$

We seek a local expression R whose evolution can be written as $R_t + Q_x = 0$. Wherever $R > 0$, this law defines a conservation coordinate,

$$d\xi = R\,dx - Q\,dt, \qquad x_\xi = R^{-1}, \qquad V = x_t\big|_\xi = Q/R. \tag{18}$$

Equal coordinate increments carry equal amounts of R, so larger values assign smaller physical cells. The corresponding flux determines how the nodes move. Selecting a conservation law thus specifies both the initial distribution and an evolution rule for the grid.

The search can be organized using the derivative $D = \partial_t - u\partial_x$, which follows the velocity $-u$. For any candidate density,

$$R_t + Q_x = 0 \quad \Longleftrightarrow \quad DR - u_x R = -\partial_x(Q + uR).$$

We therefore compute $DR - u_x R$ and look for a spatial derivative in the remaining terms. The following constructions use simple expressions suggested by the transport equations. The conservation condition determines the necessary powers and relative coefficients; background normalization then makes the coordinate densities comparable.

Physical mass

The density equation $\rho_t = (u\rho)_x$ already has the required divergence form. It gives

$$R_\rho = \rho, \qquad Q_\rho = -u\rho, \qquad V_\rho = -u. \tag{19}$$

No additional combination of fields is needed. For a cell moving with endpoint velocities $-u$, the boundary transport cancels the change in its integrated physical density. The conserved mass labels the cell and determines its width through (18). This recovers the reciprocal construction of Section 2.1 and provides the structural baseline for the other choices. In a density depression, equal mass requires a larger cell, explaining the reduced concentration around the present velocity peaks.

Balancing curvature and mass stretching

To identify another density, return to the differentiated velocity equation. With $m = u_{xx} + 2$, the two transport relations are

$$Dm = 2u_x m + c^2 \rho \rho_x, \qquad D\rho = u_x \rho. \tag{20}$$

The curvature variable has two copies of the stretching term, while mass has one. A conservation density needs one copy, as the criterion above shows. This suggests dividing m by a power of ρ. Rather than fixing the power in advance, set $R = m\rho^{-\alpha}$. Direct differentiation yields

$$DR - u_x R = (1-\alpha) u_x m \rho^{-\alpha} + c^2 \rho^{1-\alpha} \rho_x.$$

Choosing $\alpha = 1$ removes the excess stretching. The remaining term is $c^2 \rho_x$, which is already a spatial derivative. Since $m = 2$ and $\rho = 1$ at the background, we normalize the density by a factor two and choose a flux that vanishes there:

$$R_m = \frac{m}{2\rho}, \qquad Q_m = -uR_m - \frac{c^2}{2}(\rho - 1), \qquad (R_m)_t + (Q_m)_x = 0. \tag{21}$$

The ratio consequently arises from the different transport weights of curvature and mass. Its allocation responds to their combination. For the one-soliton family, let $s = 1 - 2/(cp)$ and $z = \tanh(\theta_1/2)$. Then $R_m = 1 + s^2(1-z^2)(1+s^2z^2)/(1-s^2z^2)^2 \geq 1$ on the smooth branch. This density increases within the wave, reversing the allocation produced by physical mass. The numerical calculation evaluates the same expression for both wave families and checks its positivity during evolution.

A quadratic balance for slope and density

For a third choice, we use the velocity gradient itself. Writing $w = u_x$, the original integrated equation and the mass equation give

$$Dw = \frac{1}{2}w^2 + 4u + \frac{c^2}{2}(\rho^2 - 1), \qquad D\rho = \rho w. \tag{22}$$

These relations describe the local rates of w and ρ. They are a convenient starting point for a low-order balance because the equation specifies u_{tx}, whereas obtaining u_t requires a spatial integration and its boundary value. A density built from w and ρ can therefore be differentiated locally without first reconstructing u_t.

We look for a quadratic expression that records the size of the velocity slope and the density contribution already present in (22). The term w^2 records slope magnitude independently of its sign. The combination $\rho^2 - 1$ is selected by the density coupling in the slope equation and vanishes at the unit background. Starting with these two terms, write $E_A = w^2 + A(\rho^2 - 1)$, leaving the relative coefficient A undetermined. Substitution into (22) gives

$$DE_A - wE_A = 8uw + (c^2 + A)\rho^2 w + (A - c^2)w. \tag{22a}$$

The terms involving only uw and w are spatial derivatives, since $w = u_x$. The mixed term $\rho^2 w$ is the remaining exchange between the two fields. Setting $A = -c^2$ cancels it and leaves $8uu_x - 2c^2 u_x = \partial_x(4u^2 - 2c^2 u)$. Thus the equations determine the relative sign and coefficient in

$$E = u_x^2 - c^2(\rho^2 - 1), \qquad E_t + (Q_E)_x = 0, \qquad Q_E = -uE - 4u^2 + 2c^2 u. \tag{22b}$$

The cancellation identifies a conserved quadratic balance between slope and density. We call it energy-type to describe this balance in the present sign convention. Its sign depends on the fields; the use of a positive density for mesh generation is a separate step.

Because E vanishes at the background, we add a uniform baseline and use the coupling scale c^2 to set the relative size of its contribution. Our mesh choice is

$$R_E = 1 + \frac{E}{c^2} = 2 - \rho^2 + \frac{u_x^2}{c^2}, \qquad Q_{R_E} = \frac{Q_E}{c^2}, \qquad (R_E)_t + (Q_{R_E})_x = 0. \tag{23}$$

Adding a constant preserves the local conservation law. Here it gives $R_E = 1$ and $Q_{R_E} = 0$ at the background, matching the normalization of the other two coordinates. This is a specified mesh-design convention: the conservation law fixes the quadratic combination, while its scale relative to the uniform baseline controls the allocation. We use (23) unchanged in both cases and at each resolution.

For a positive density depression with $\rho \leq 1$, the expression in (23) satisfies $R_E \geq 1$. Both a larger slope magnitude and a deeper density depression increase its value. Its geometry can therefore be interpreted directly from the physical profiles. The curvature–mass coordinate weights $u_{xx} + 2$ relative to ρ; the energy-type coordinate instead weights slope magnitude and the density deficit. Their numerical comparison asks how these two ways of identifying wave structure affect the coupled field errors.

5.2. Common field evolution and mesh implementation

We initialize each moving grid by equidistributing its exact initial density on the physical interval $[-4, 4]$:

$$\int_{-4}^{x_k(0)} R(x,0)\,dx = \frac{k}{N-1}\int_{-4}^{4} R(x,0)\,dx, \qquad k = 0, \ldots, N-1. \tag{24}$$

At each RK4 stage, the interior nodes follow $\dot{x}_k = Q_k/R_k$, evaluated from the current numerical fields. Both endpoint positions are fixed and receive exact time-dependent field values. The conservation-coordinate velocity applies to the interior; the endpoint closure defines the finite-domain computation. The uniform fixed grid uses $V = 0$.

All four calculations evolve the same equations along their nodes:

$$\dot{m} = (u+V)m_x + 2u_x m + c^2\rho\rho_x, \qquad \dot{\rho} = (u+V)\rho_x + \rho u_x, \qquad u_{xx} = m - 2. \tag{25}$$

We use three-point differences on the current mesh and recover u from the discrete boundary-value problem. Initially, $m_k = \mathcal{D}_2 u_*(x_k, 0) + 2$, so this recovery reproduces the sampled exact velocity; ρ_k is sampled from the exact density. The nonuniform operators, their consistency expansions and the tridiagonal recovery are given in Appendix B. The velocity gradient in R_E is evaluated by the same three-point first-derivative operator. The monitor and its flux are reconstructed at every stage from these numerical fields. Both physical fields are nodal quantities. Initial accumulated-mass inversion determines the node locations, while the evolved errors are evaluated directly at those locations.

5.3. Numerical comparison

We use the two wave families of Section 4.1, with $c = 1$, $N = 2000$ actual nodes and RK4 step $\Delta T = 5 \times 10^{-4}$, and evolve to $T = 0.25$. The fixed-grid baseline and all three SAMM calculations share the domain, field stencils and boundary data. At the terminal time, we evaluate the continuous exact fields at each method's own nodes and take the maximum errors over $[-2.5, 2.5]$. Tables 7 and 8 give these errors and their ratios to fixed FD; the figures locate them relative to the cell-width distribution. The comparisons measure the combined effect of initial allocation and subsequent node motion.

One-soliton case

The single wave makes the contrasting mesh geometries particularly clear. Physical mass decreases inside the wave, whereas R_m and R_E increase there. Table 7 records the resulting terminal field errors. Figure 9 shows the cell widths and the error profiles on the same physical interval.

Table 7. One soliton: nodal errors at $T = 0.25$ with 2000 nodes. Ratios are relative to fixed FD. Bold entries mark the smallest error in each field.

Mesh	E_u	E_ρ	E_u/E_u^{fixed}	$E_\rho/E_\rho^{\text{fixed}}$
Fixed FD	$\mathbf{3.059 \times 10^{-8}}$	2.615×10^{-7}	1.000	1.000
R_ρ	3.016×10^{-7}	5.501×10^{-7}	9.859	2.104
R_m	4.501×10^{-8}	1.535×10^{-7}	1.471	0.587
R_E	5.009×10^{-8}	$\mathbf{1.329 \times 10^{-7}}$	1.638	0.508

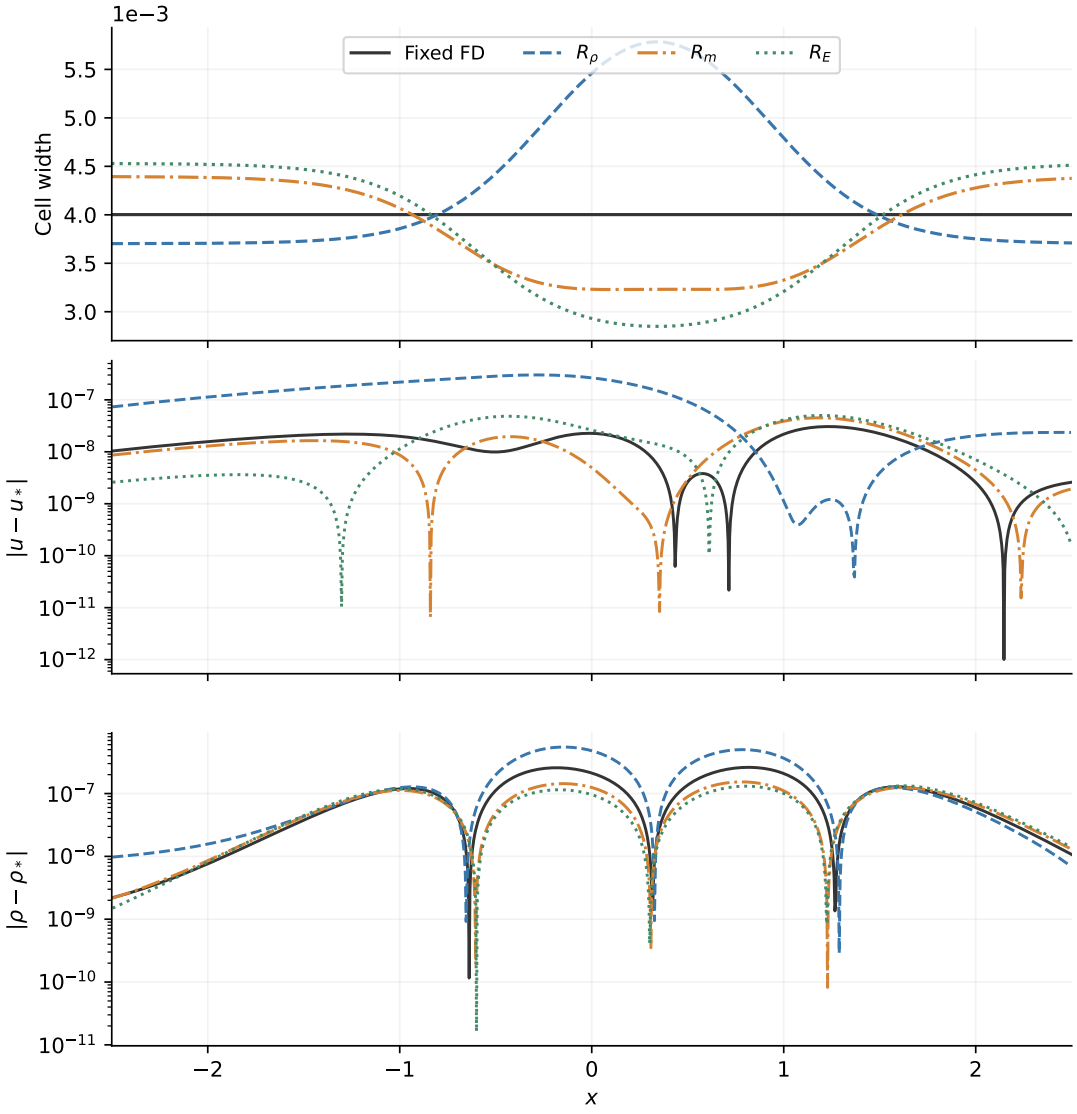


Figure 9. One-soliton SAMM comparison at $T = 0.25$. Top: interior cell widths. Middle and bottom: absolute nodal field errors on logarithmic scales. Curves connect actual samples; the plotted window is $[-2.5, 2.5]$.

The mass-based mesh has larger errors in both fields than fixed FD. Its velocity error is about 9.86 times the fixed-grid value and its density error is about 2.10 times that value. Both alternative coordinates improve the density field: the error ratios are approximately 0.587 for R_m and 0.508 for R_E. The energy-type coordinate therefore gives the smallest density error in this case. Its velocity error is about 1.64 times the fixed-grid value, compared with 1.47 for R_m. The density depression and slope contribution in (23) supply a useful allocation for ρ, while the velocity recovery and coupled evolution respond differently to that allocation.

To examine the role of the initial mesh, we also apply the velocity recovery separately from time evolution. At 1000 nodes, supplying exact sampled curvature to the Poisson solver gives single-wave velocity

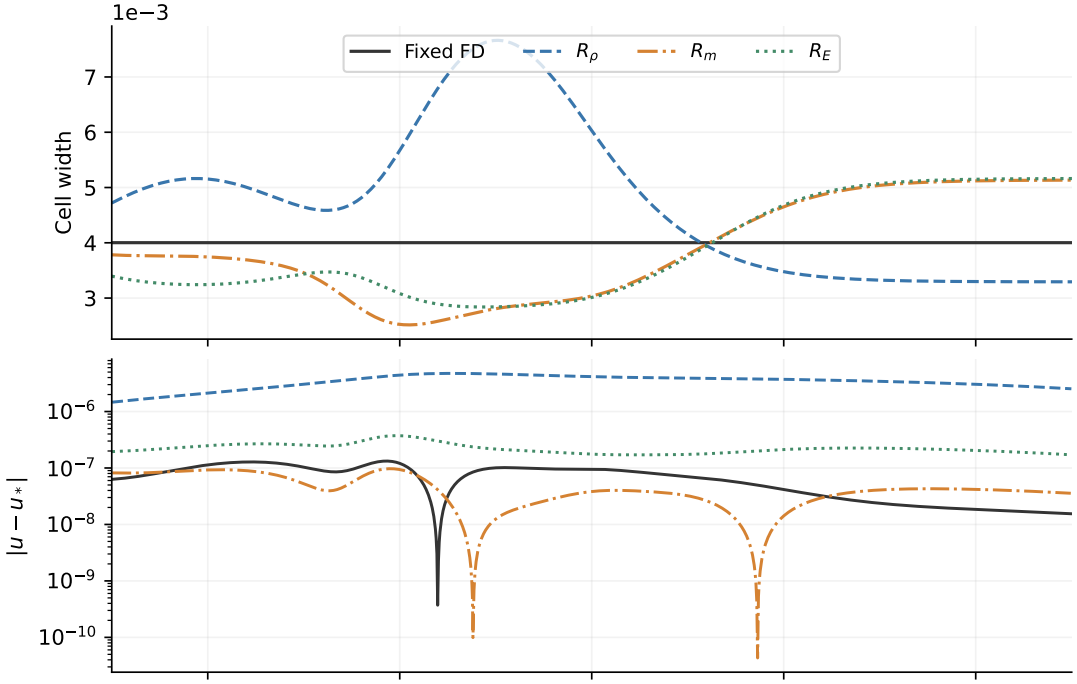
errors of approximately 1.39×10^{-6}, 9.46×10^{-7} and 2.00×10^{-6} on the fixed, R_m and R_E grids, respectively. The curvature–mass grid improves this isolated recovery even though its terminal velocity error is larger than the fixed-grid error. The energy-type grid has a larger error in both measurements. These results separate two contributions worth examining: the spatial recovery on a given grid and the curvature error accumulated during evolution. The diagnostic supplies analytic curvature, while the evolution uses the consistent discrete initialization in Section 5.2. Appendix C.3 gives the differentiation and recovery errors for all four grids and both profiles.

Two-soliton case

The two-wave profile tests the same density choices in a less uniform spatial structure. Table 8 reports the terminal errors, and Figure 10 relates their locations to the mesh allocation around the two waves.

Table 8. Two solitons: nodal errors at $T = 0.25$ with 2000 nodes. Ratios are relative to fixed FD. Bold entries mark the smallest error in each field.

Mesh	E_u	E_ρ	E_u/E_u^{fixed}	$E_\rho/E_\rho^{\text{fixed}}$
Fixed FD	1.328×10^{-7}	6.208×10^{-7}	1.000	1.000
R_ρ	4.738×10^{-6}	2.316×10^{-6}	35.686	3.730
R_m	$\mathbf{9.720 \times 10^{-8}}$	$\mathbf{3.242 \times 10^{-7}}$	0.732	0.522
R_E	3.741×10^{-7}	4.674×10^{-7}	2.818	0.753



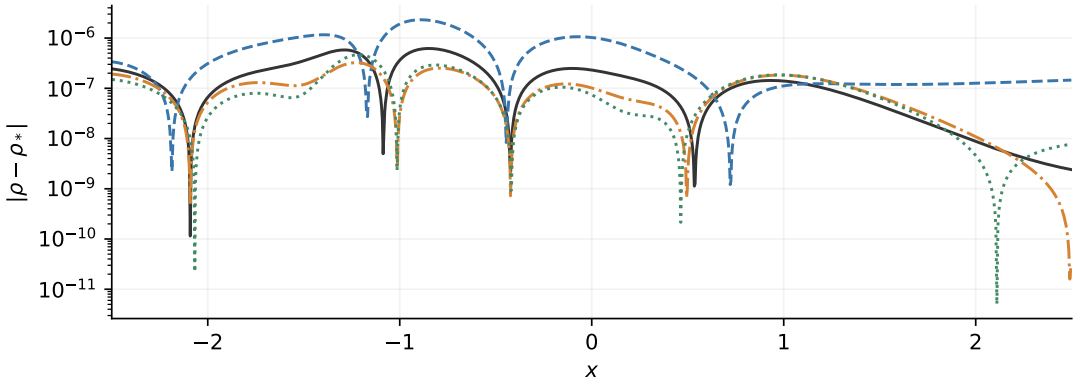


Figure 10. Two-soliton SAMM comparison at $T = 0.25$. Top: interior cell widths. Middle and bottom: absolute nodal field errors on logarithmic scales. Curves connect actual samples; the plotted window is $[-2.5, 2.5]$.

For the two-wave profile, R_m improves both fields, with velocity and density errors approximately 0.732 and 0.522 times the fixed-grid values. The energy-type coordinate reduces the density error to about 0.753 of the baseline, while its velocity error is about 2.82 times larger. Physical mass gives the largest errors in both fields. Thus both alternatives direct resolution toward wave structure in a way that benefits the density, but the curvature–mass ratio also improves the recovered velocity for this profile. The relative error curves show where these differences occur across the two wave regions.

All three densities, the physical density and the mesh widths remain positive at the sampled full-step states; the smallest recorded cell width is 0.0025. Halving the time step changes the terminal node positions and field values by at most 3.137×10^{-13}. These checks accompany the terminal comparisons for all four mesh choices; the full values are reported in Appendix C.2.

Upon refinement from 1000 to 2000 nodes, the observed orders for the four mesh choices and both fields lie between 1.999 and 2.001. The errors and observed orders are listed in Appendix C.1. Their relative errors remain close across these resolutions. The density choice primarily changes the error coefficients in this regime, while the common field discretization retains its second-order trend.

Across the two profiles, R_m gives the smallest velocity errors among the three moving meshes and improves density accuracy relative to fixed FD. The energy-type coordinate gives the smallest density error for the single wave, while R_m gives the smaller density error for the two-wave profile. Their different rankings connect the choice of local field information to a measurable change in accuracy. The quadratic balance emphasizes slope magnitude and density deficit; the ratio $m/(2\rho)$ combines curvature with the density that stretches the grid. The results identify the latter as the more balanced allocation in these tests, and show how the energy-type allocation can favor one component of a smooth wave.

6. Conclusions

We have studied how the analytical structure of the two-component Hunter–Saxton system can guide its numerical approximation. The existing integrable semi-discretization provides exact finite-lattice trajectories, and the reciprocal transformation connects those trajectories to physical coordinates and cell densities. Using these relations, we have compared spatial formulations, isolated time-integration errors and examined the allocation of moving nodes within a common numerical study.

The spatial results have a concrete explanation in the equations. The difference between the integrable and direct moving-grid vector fields contains the first-order term $ac(1 - \rho^2)$, while the exact lattice phase has a corresponding first-order departure from its continuous limit. The observed errors decrease approximately linearly for integrable SD and quadratically for the direct finite-difference formulations. This difference in order accounts for their distinct refinement trends in the selected smooth soliton tests. The exact solutions make the distinction between these spatial effects and temporal integration error measurable, allowing the time resolution to be chosen with the dominant error source in view.

The conservation laws provide a second route from structure to computation. We obtain three mesh

densities from the field equations through physical mass conservation, a balance of curvature and mass stretching, and cancellation of slope–density exchange in a quadratic expression. Giving the construction explicitly makes the allocation rule traceable to the quantities represented by each density. In the tests, physical mass expands the mesh near the velocity peaks, whereas the curvature–mass and energy-type densities concentrate resolution there. The curvature–mass ratio improves density accuracy in both cases and velocity accuracy in the two-soliton case. The energy-type choice gives the smallest single-wave density error, accompanied by a larger velocity error. The common second-order refinement trend persists while the fieldwise error coefficients vary with the chosen coordinate. The initial recovery diagnostic further shows why grid selection needs to be assessed together with the accumulated field evolution. Among the three moving meshes, the curvature–mass ratio gives the most balanced results for the present profiles.

The perturbation experiments complement these accuracy comparisons by following the response within each numerical formulation. An initial velocity disturbance generates a density response through the coupled evolution, and the measured histories remain consistent under amplitude and time-step refinement. Taken together, the results connect exact solution structure with the selection of reference quantities, the interpretation of numerical error and the design of mesh resolution. This offers a practical way to use the analytical information carried by an integrable model throughout a numerical computation.

Appendix A. The correction in the moving-grid slope equation

We expand the change of variables used in Section 3.1 to obtain the difference between the two moving-grid vector fields. The calculation compares both right-hand sides at the same state, with positive cell width d_k and fixed reciprocal spacing a. Write $v_k = d_k w_k$. Since $\dot{d}_k = -v_k$, differentiation gives

$$\dot{w}_k = \frac{\dot{v}_k}{d_k} - \frac{v_k \dot{d}_k}{d_k^2} = \frac{\dot{v}_k}{d_k} + w_k^2. \tag{A.1}$$

Substituting (8) into (A.1), the quadratic velocity term combines with the last term to give

$$F_{\mathrm{SD}}^{(w)} = 2(u_k + u_{k-1}) + \frac{1}{2} w_k^2 + \frac{1}{2}\left(d_k - \frac{\alpha}{d_k}\right)^2 - \frac{c^2}{2}. \tag{A.2}$$

The direct moving-grid approximation (9) has the same adjacent-node velocity mean and the same quadratic slope contribution. Subtraction therefore leaves

$$F_{\mathrm{SD}}^{(w)} - F_{\text{moving FD}}^{(w)} = \frac{1}{2}\left[\left(d_k - \frac{\alpha}{d_k}\right)^2 - c^2 \rho_k^2\right]. \tag{A.3}$$

Now use $d_k = a/\rho_k$ and $\alpha = a(a-c)$. The expression inside the square becomes

$$d_k - \frac{\alpha}{d_k} = c\rho_k + a(\rho_k^{-1} - \rho_k). \tag{A.4}$$

Expanding (A.3) proves the identity

$$F_{\mathrm{SD}}^{(w)} - F_{\text{moving FD}}^{(w)} = ac(1-\rho_k^2) + \frac{a^2}{2}(\rho_k^{-1} - \rho_k)^2. \tag{A.5}$$

For states with ρ_k bounded above and bounded away from zero, the final term is uniformly $O(a^2)$. The leading correction is linear in a and vanishes at $\rho_k = 1$. Its spatial variation is thus linked to the density profile. In the smooth depression waves used here, the leading term is positive for $c > 0$ wherever $0 < \rho_k < 1$.

The exact lattice phase provides a second view of the finite-spacing dependence. For fixed p_i, q_i and sufficiently small a, Taylor expansion of (6) gives

$$\frac{\sigma_i(a)}{a} = (p_i - q_i) + \frac{a}{2}(p_i^2 - q_i^2) + \frac{a^2}{3}(p_i^3 - q_i^3) + O(a^3). \tag{A.6}$$

At a fixed reciprocal position $X = ka$, the lattice spatial phase is consequently $(p_i - q_i)X + \frac{1}{2}a(p_i^2 - q_i^2)X + O(a^2)$. This is a first-order perturbation of the continuous phase on a bounded X-interval. Equations (A.5) and (A.6) identify first-order contributions in the evolution law and in its exact wave family, respectively. Their coefficients refer to different quantities; the physical-field errors in Section 4 are measured after reconstruction and evolution.

Appendix B. Nonuniform differences and velocity recovery

The SAMM calculations use the same three-point field operators on each evolving grid. We record these operators and their consistency terms to specify how the local mesh geometry enters the computation. At an interior node x_j, let $h_- = x_j - x_{j-1} > 0$ and $h_+ = x_{j+1} - x_j > 0$. Differentiating the quadratic interpolant through these three nodes gives

$$\mathcal{D}_1 f_j = -\frac{h_+}{h_-(h_- + h_+)} f_{j-1} + \frac{h_+ - h_-}{h_- h_+} f_j + \frac{h_-}{h_+(h_- + h_+)} f_{j+1}, \tag{B.1}$$

$$\mathcal{D}_2 f_j = \frac{2}{h_- + h_+} \left(\frac{f_{j+1} - f_j}{h_+} - \frac{f_j - f_{j-1}}{h_-} \right). \tag{B.2}$$

For equal adjacent widths these are the centered uniform-grid formulas. The derivatives in the field equations, including the velocity gradient entering R_E, are evaluated on the current node positions at each time stage.

B.1. Local consistency and mesh smoothness

Taylor expansion about x_j, for a function with bounded derivatives through order five in the neighboring cells, yields

$$\mathcal{D}_1 f_j - f_x(x_j) = \frac{h_- h_+}{6} f_{xxx}(x_j) + \frac{h_- h_+ (h_+ - h_-)}{24} f_{xxxx}(x_j) + O(h_{\max}^4), \tag{B.3}$$

$$\mathcal{D}_2 f_j - f_{xx}(x_j) = \frac{h_+ - h_-}{3} f_{xxx}(x_j) + \frac{h_+^2 - h_+ h_- + h_-^2}{12} f_{xxxx}(x_j) + O(h_{\max}^3), \tag{B.4}$$

where $h_{\max} = \max(h_-, h_+)$. The second-derivative formula depends on both the local cell size and the difference between neighboring sizes. This explains why a node-allocation rule must be considered together with the smoothness of the resulting mesh.

To make the scaling explicit, suppose $x_j = x(\xi_j)$ with uniform label spacing ε, a smooth mapping with bounded derivatives, and x_ξ bounded away from zero. Then

$$h_\pm = x_\xi \varepsilon \pm \frac{1}{2} x_{\xi\xi} \varepsilon^2 + \frac{1}{6} x_{\xi\xi\xi} \varepsilon^3 + O(\varepsilon^4), \qquad h_+ - h_- = x_{\xi\xi} \varepsilon^2 + O(\varepsilon^4). \tag{B.5}$$

Both leading contributions in (B.4) are therefore $O(\varepsilon^2)$. For a smooth positive conservation density, $x_\xi = 1/R$ and $x_{\xi\xi} = -R_x/R^3$. These relations connect the consistency coefficients to the local density and its gradient. They describe the spatial operator on a smooth mapped grid; the numerical refinements in Section 5 measure the complete coupled evolution, including its endpoint treatment.

B.2. The discrete boundary-value problem

Let $x_0 = x_L$ and $x_{N-1} = x_R$, with prescribed values u_L and u_R. The interior recovery $\mathcal{D}_2 u = m - 2$ has the tridiagonal form

$$A_j^- u_{j-1} + A_j^0 u_j + A_j^+ u_{j+1} = m_j - 2, \qquad A_j^- = \frac{2}{h_-(h_- + h_+)}, \quad A_j^0 = -\frac{2}{h_- h_+}, \quad A_j^+ = \frac{2}{h_+(h_- + h_+)}. \tag{B.6}$$

For the first interior row, the term $A_1^- u_L$ is moved to the right-hand side; for the last row, the term $A_{N-2}^+ u_R$ is moved in the same way. The remaining interior system determines the velocity used in the stage right-hand side. Positive cell widths make both off-diagonal coefficients positive and each full row sum zero, yielding the discrete maximum principle with Dirichlet boundary data.

The same principle gives a useful bound for the isolated recovery test. Let u_* be the exact velocity and set $m_j = u_*''(x_j) + 2$, with exact endpoint values. Write $\tau_j = \mathcal{D}_2 u_*(x_j) - u_*''(x_j)$. The recovery error $e_j = u_j - u_*(x_j)$ satisfies $\mathcal{D}_2 e_j = -\tau_j$ and vanishes at both endpoints. The quadratic barrier $b(x) = (x - x_L)(x_R - x)/2$ is represented exactly by (B.2), so $\mathcal{D}_2 b = -1$. Applying the maximum principle to $e \pm \|\tau\|_\infty b$ gives

$$|e_j| \le \|\tau\|_\infty \frac{(x_j - x_L)(x_R - x_j)}{2}, \qquad \|e\|_\infty \le \frac{(x_R - x_L)^2}{8} \|\tau\|_\infty. \tag{B.7}$$

The norms in this bound cover all interior nodes. If the input curvature has an error δm, the same argument replaces $\|\tau\|_\infty$ by $\|\delta m - \tau\|_\infty$. Thus velocity accuracy depends on the curvature delivered by the evolution as well as on the recovery operator. In the actual time-dependent experiments, the initialization $m_j = \mathcal{D}_2 u_*(x_j, 0) + 2$ makes the initial velocity recovery exact up to the linear-solve accuracy. The diagnostic in Appendix C instead supplies the analytic curvature to isolate the spatial recovery error.

Appendix C. Supplementary refinement and recovery checks

We collect the numerical checks underlying the SAMM comparisons. The two cases use $p_1 = 5$ and $(p_1, p_2) = (1.1, 1.25)$, respectively, with $c = 1$ and the boundary and initialization conventions of Section 5.2. All field errors below are measured directly at the computational nodes in $[-2.5, 2.5]$. The grid has N nodes and $N - 1$ intervals.

C.1. Spatial refinement

Table C.1 compares the terminal errors at $N = 1000$ and $N = 2000$, with $T = 0.25$ and $\Delta T = 5 \times 10^{-4}$. The node counts give the nominal refinement factor $1999/999$, so we use

$$p_q = \frac{\log(E_q^{(1000)} / E_q^{(2000)})}{\log(1999/999)}, \qquad q \in \{u, \rho\}. \tag{C.1}$$

The moving meshes are initialized from the same conserved density at each resolution. The resulting orders describe refinement of each complete mesh formulation at a common node budget.

Table C.1. Nodal refinement at $T = 0.25$ for fixed FD and the three SAMM formulations.

Mesh	$E_u^{(1000)}$	$E_u^{(2000)}$	p_u	$E_\rho^{(1000)}$	$E_\rho^{(2000)}$	p_ρ
One soliton						
Fixed FD	1.225×10^{-7}	3.059×10^{-8}	2.000	1.047×10^{-6}	2.615×10^{-7}	2.000
R_ρ	1.207×10^{-6}	3.016×10^{-7}	2.000	2.202×10^{-6}	5.501×10^{-7}	2.000
R_m	1.802×10^{-7}	4.501×10^{-8}	2.000	6.146×10^{-7}	1.535×10^{-7}	2.000
R_E	2.005×10^{-7}	5.009×10^{-8}	2.000	5.320×10^{-7}	1.329×10^{-7}	2.000
Two solitons						
Fixed FD	5.316×10^{-7}	1.328×10^{-7}	2.000	2.485×10^{-6}	6.208×10^{-7}	2.000
R_ρ	1.897×10^{-5}	4.738×10^{-6}	2.000	9.267×10^{-6}	2.316×10^{-6}	1.999
R_m	3.893×10^{-7}	9.720×10^{-8}	2.001	1.298×10^{-6}	3.242×10^{-7}	2.000
R_E	1.498×10^{-6}	3.741×10^{-7}	2.000	1.871×10^{-6}	4.674×10^{-7}	2.000

The errors decrease approximately quadratically for all four choices. Their relative sizes distinguish the density choices within that common trend. In particular, the smaller density error of the curvature–mass

and energy-type meshes persists under refinement, while the velocity ordering remains dependent on the wave profile and the selected coordinate.

C.2. Time-step and positivity checks

For each $N = 2000$ calculation, we repeat the evolution with half the RK4 step. At the common terminal time, define the change at matching node labels by

$$\delta_{\Delta T} = \max\left\{\|x_{\Delta T} - x_{\Delta T/2}\|_\infty, \|u_{\Delta T} - u_{\Delta T/2}\|_\infty, \|\rho_{\Delta T} - \rho_{\Delta T/2}\|_\infty\right\}. \tag{C.2}$$

Table C.2 reports this change, the minimum cell width, and the minimum physical and monitor densities sampled at full time steps of the baseline run. The minimum monitor density during the evolution is taken over interior nodes, where the conservation-coordinate velocity is imposed; the exact initial monitor is also included in the check. For fixed FD, $R = 1$.

Table C.2. Step-halving changes and sampled minima for the 2000-node experiments.

Mesh	$\delta_{\Delta T}$	$\min h$	$\min \rho$	$\min R$
One soliton				
Fixed FD	4.663×10^{-15}	4.002×10^{-3}	0.640000	1.000000
R_ρ	3.790×10^{-14}	3.702×10^{-3}	0.640000	0.640000
R_m	1.240×10^{-13}	3.229×10^{-3}	0.640000	1.000001
R_E	1.205×10^{-13}	2.849×10^{-3}	0.640000	1.000001
Two solitons				
Fixed FD	4.330×10^{-15}	4.002×10^{-3}	0.424151	1.000000
R_ρ	1.986×10^{-13}	3.092×10^{-3}	0.424152	0.424151
R_m	3.137×10^{-13}	2.514×10^{-3}	0.424151	1.000001
R_E	8.208×10^{-14}	2.830×10^{-3}	0.424151	1.000001

The step-halving changes are several orders of magnitude below the field errors in Table C.1. The recorded positive densities and cell widths show that the tested evolutions stay within the sampled admissible mesh regime. Together, these checks support interpreting the differences between the terminal errors as spatial effects in the stated experiments.

C.3. Recovery on the initial grids

We also examine the velocity recovery before time integration. On each $N = 1000$ initial grid, we sample the exact velocity and analytic curvature, compute $\mathcal{D}_2 u_* - u_*''$, and solve the discrete Poisson problem with that analytic curvature and exact endpoint data. Table C.3 reports the maximum differentiation defect and recovered velocity error in the observation interval. It also gives the ratio of the largest to the smallest cell over the full physical domain.

Table C.3. Initial-grid differentiation and isolated velocity recovery at 1000 nodes.

Mesh	$\max \lvert\mathcal{D}_2 u_* - u_*''\rvert$	$\max \lvert u_{\mathrm{rec}} - u_*\rvert$	$h_{\max}/h_{\min}$
One soliton			
Fixed FD	1.033×10^{-5}	1.385×10^{-6}	1.000
R_ρ	2.133×10^{-5}	7.853×10^{-6}	1.562
R_m	6.983×10^{-6}	9.459×10^{-7}	1.361
R_E	6.953×10^{-6}	2.000×10^{-6}	1.590
Two solitons			
Fixed FD	8.476×10^{-5}	2.872×10^{-6}	1.000
R_ρ	1.818×10^{-4}	6.096×10^{-5}	2.358
R_m	3.927×10^{-5}	5.798×10^{-6}	2.043
R_E	5.682×10^{-5}	5.372×10^{-6}	1.825

For the single wave, both alternative coordinates reduce the local differentiation defect compared with the uniform grid, but only R_m also reduces the isolated recovery error. Its larger terminal velocity error in Table C.1 therefore occurs despite the favorable initial recovery result. For the two-wave profile, both alternatives give larger isolated recovery errors than fixed FD, although R_m later produces the smallest terminal velocity error. The diagnostic and the evolved field error measure different parts of the computation. The latter includes the curvature error accumulated through transport and stretching, together with the changing mesh on which velocity is recovered.

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